

PREDICT-AML: Personalized REsponse via Drug Interaction and Cellular Traits for Acute Myeloid Leukemia

Siddhi P. Jani, Halima Bensmail, Raghvendra Mall

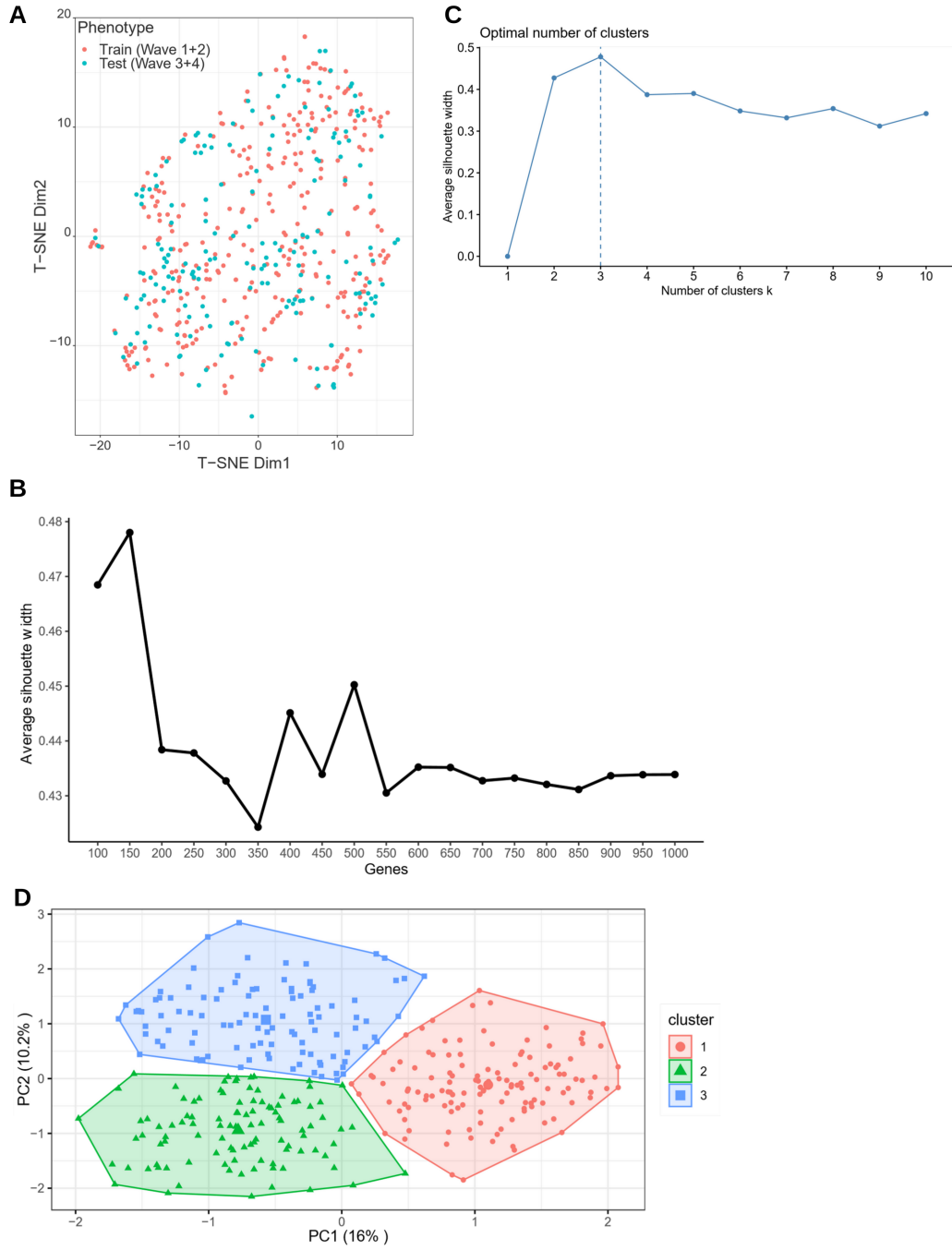


Figure 1: A. t-SNE plot showcasing uniform distribution of train and test set w.r.t. transcriptomic profiles. B. Best clustering performance obtained taking 150 most variable genes. C. Optimal number of transcriptional subgroups identified at $k = 3$. D. Distribution of the patient transcriptional profiles in the three subgroups visualized using PCA plot.

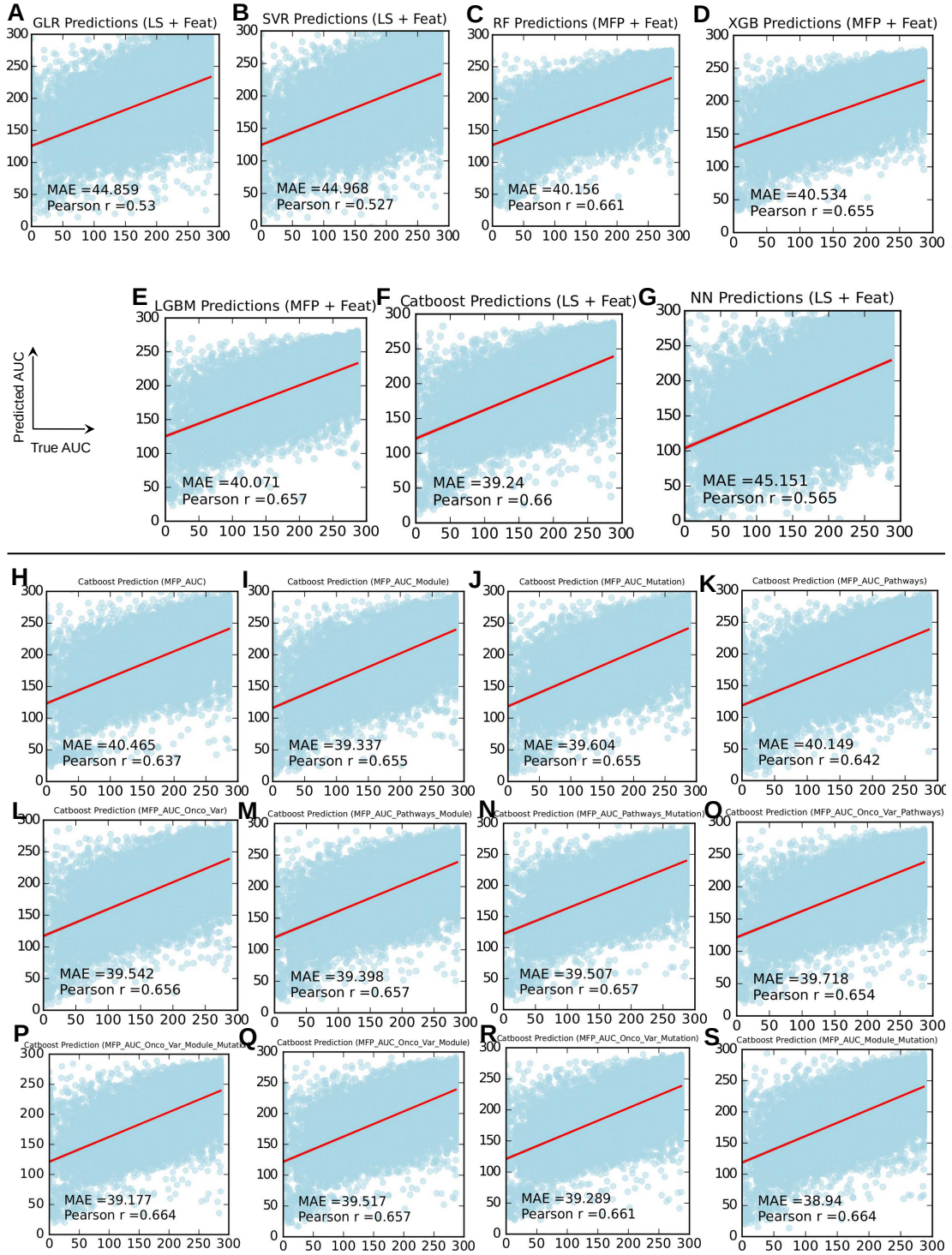


Figure 2: A-G. Performance of machine learning models on test set. H-S. Performance comparison of different combinations of patient feature sets when used as input in the CatBoost framework.

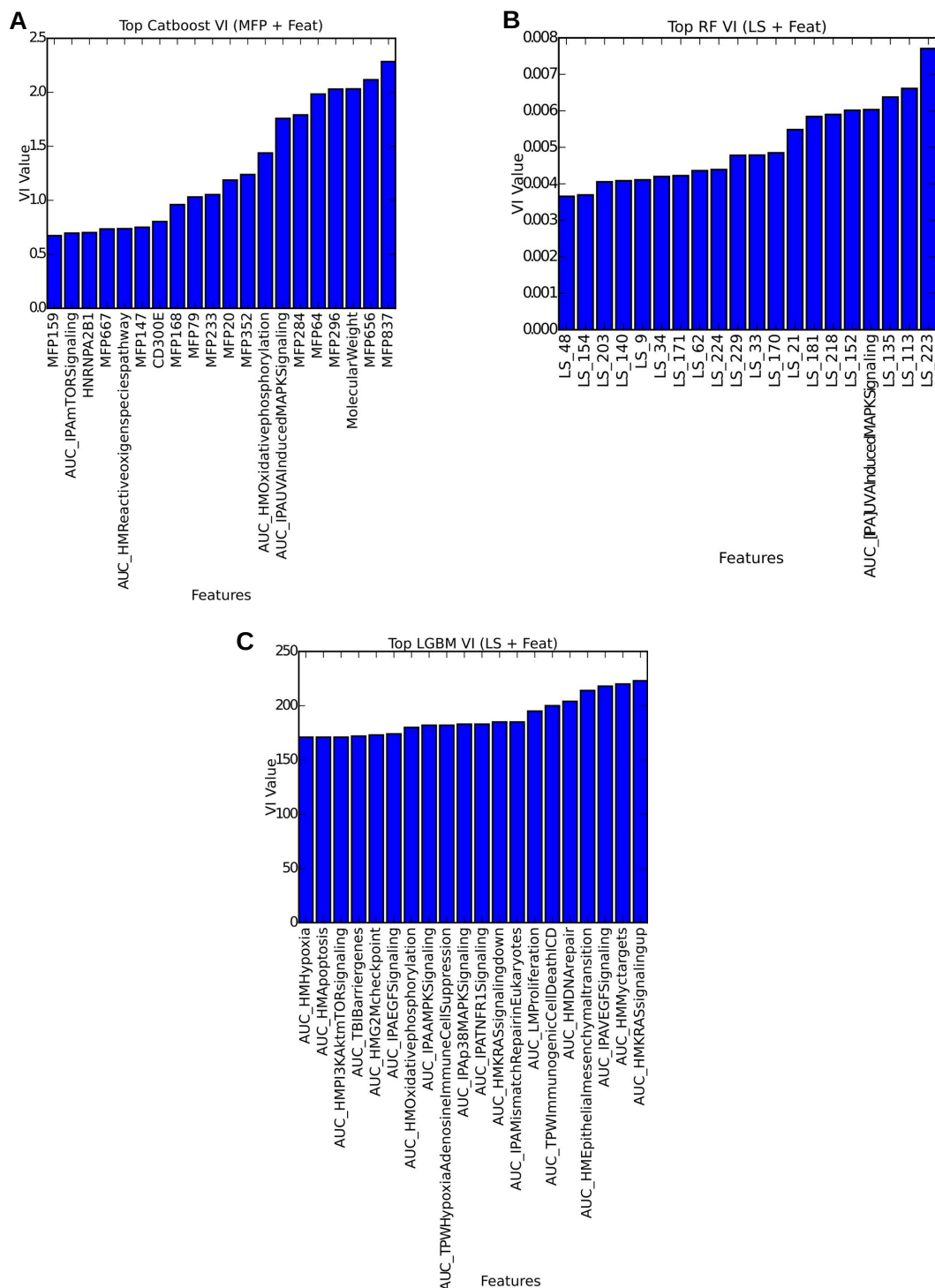
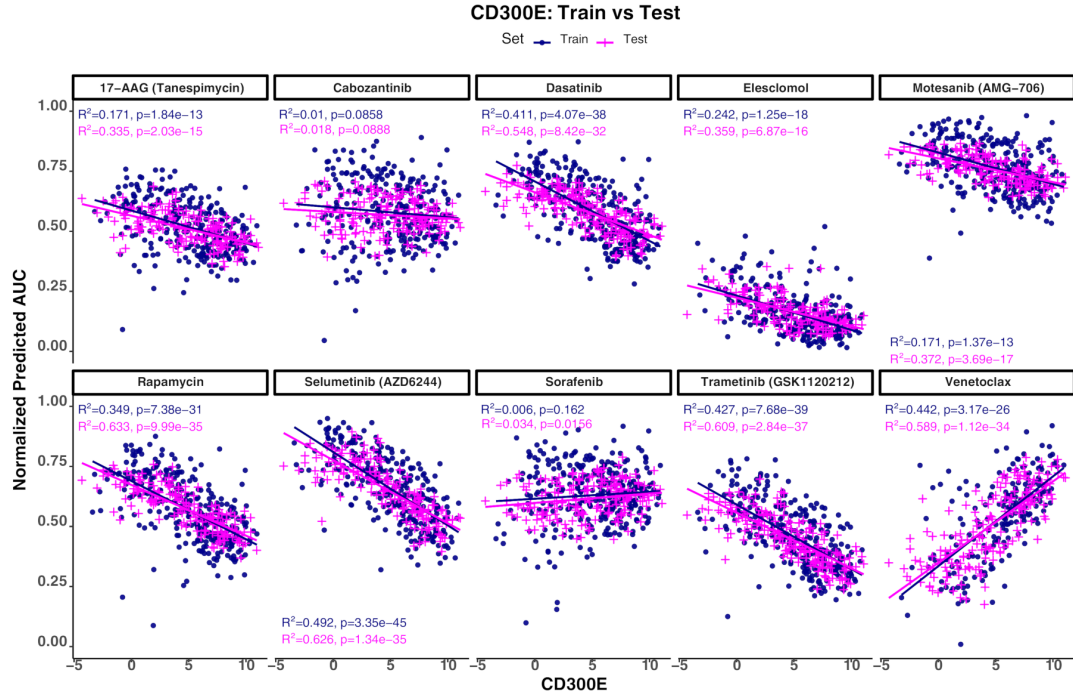
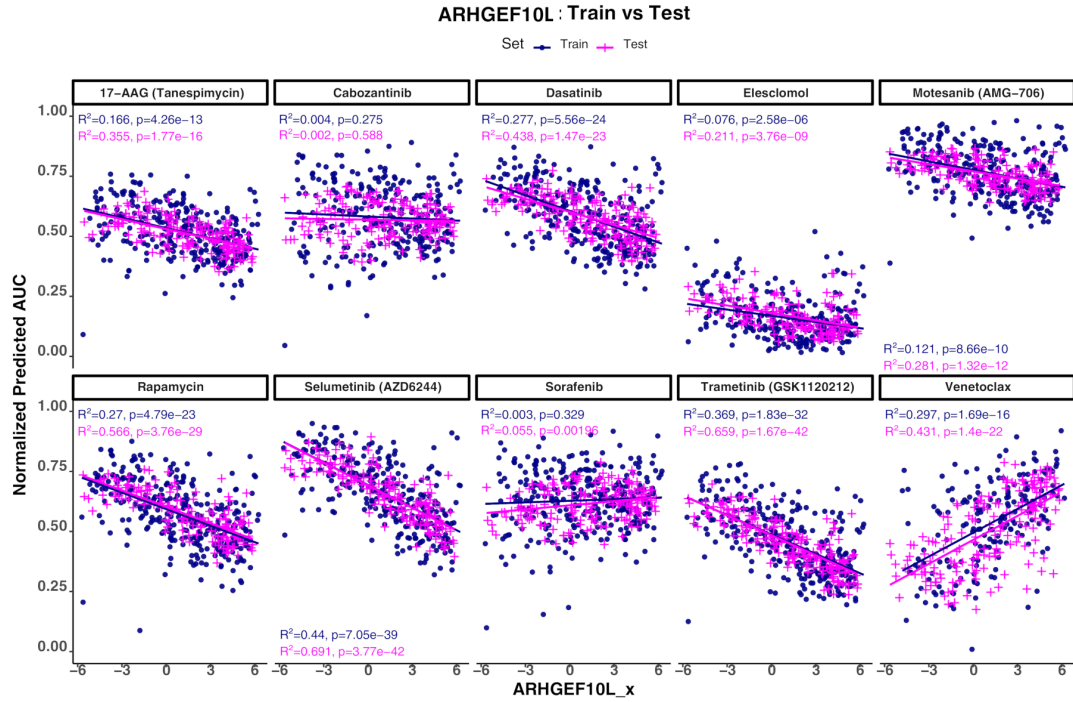


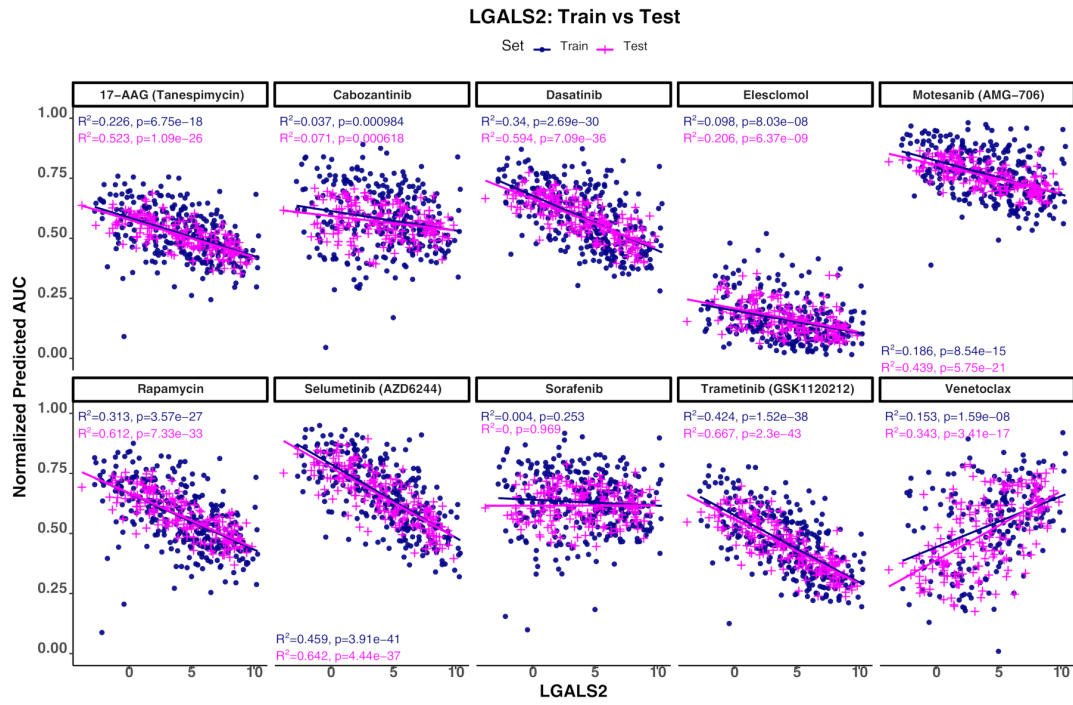
Figure 3: A-C. Top 20 most important variables of the best tree-based models, CatBoost, Random Forest (RF) and LightGBM (LGBM) are highlighted. These models use either MFP or embedding representation (LS) for drugs together with molecular feature representation (Feat) for patients.



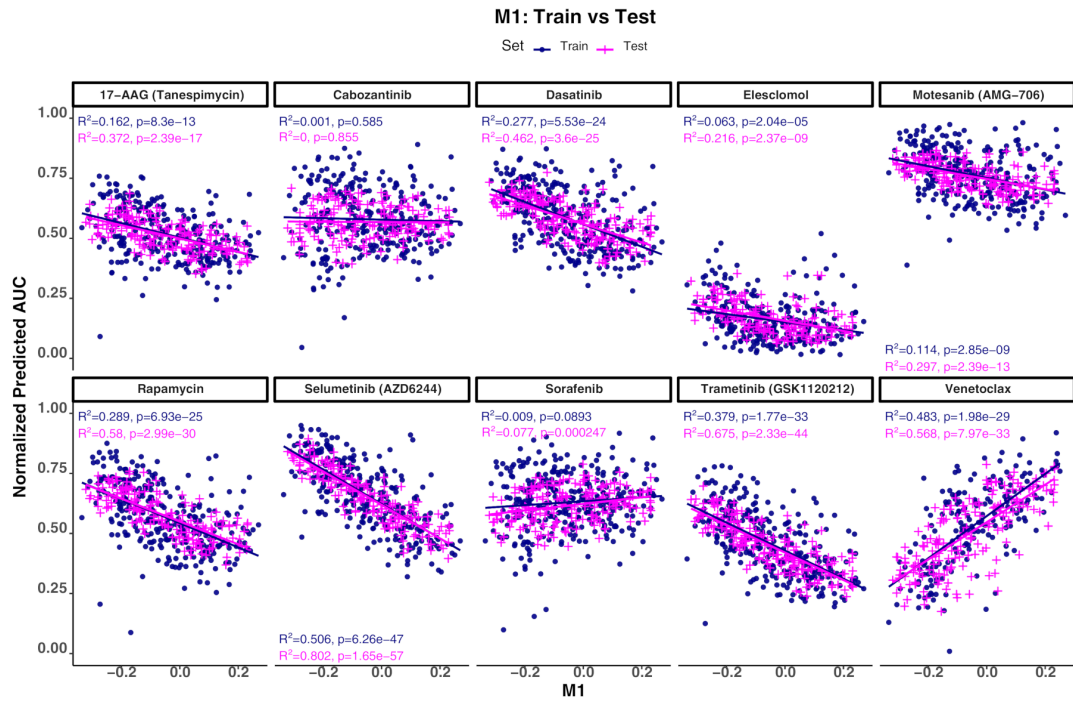
(a) Association of scaled *CD300E* expression with predicted AUC for top 10 drugs.



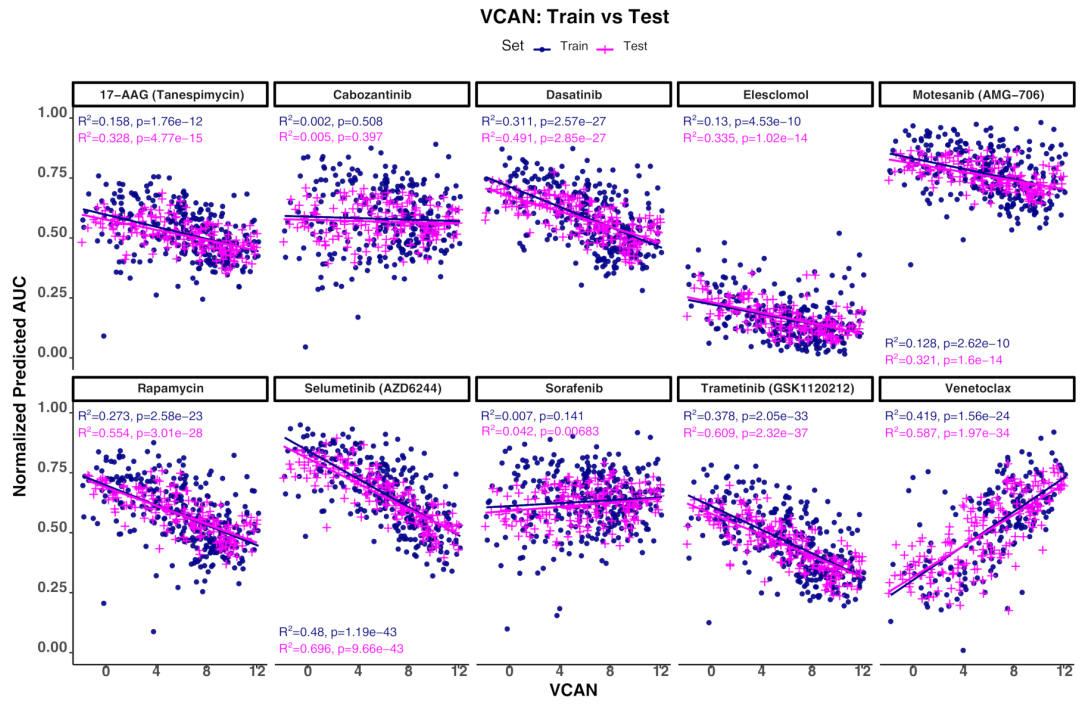
(b) Association of *ARHGEF10L* expression with predicted AUC for top 10 drugs.



(c) Association of scaled *LGALS2* expression with predicted AUC for top 10 drugs



(d) Association of scaled M1 module enrichment with predicted AUC for top 10 drugs



(e) Association of scaled *VCAN* expression with predicted AUC for top 10 drugs

Figure 4: Linear regression analysis of top features significantly associated with predicted AUC score for top 10 drugs.